

Bayesian Temporal Causal Coding (BTCC)

BTCC extends Causal Coding (CC) by integrating Bayesian uncertainty estimation and temporal latent-state modeling (inspired by Hidden Markov Neural Networks, HMNNs) to address limitations in continual learning.

Core Concept and

Traditional continual learning focuses primarily on preventing catastrophic forgetting, often neglecting crucial aspects like positive forward/backward knowledge transfer and long-term knowledge organization. CC introduces modular causal structure, but its fixed weights, deterministic gates, and noisy potency are fragile. BTCC resolves this by treating the system's weights (W_t), causal gates (G_t), and causal potency (C_t) not as fixed values, but as **probabilistic evolving latent states**. This reframes continual learning as a sequential Bayesian inference process over evolving causal structures.

Key Innovations:

BTCC introduces a dynamic, uncertainty-aware framework through:

- **Temporal Probabilistic Causal Influence:** Causal potency is modeled probabilistically, allowing the system to accumulate evidence over time.
- **Evolving Bayesian Gates:** Gates are probabilistic, enabling reusable learning pathways to dynamically strengthen, weaken, or reactivate.
- **Uncertainty-Aware Weights:** Weights are represented as probability distributions, allowing for uncertainty estimation. Highly uncertain pathways remain plastic, while confident pathways become stable, facilitating **uncertainty-driven adaptive forgetting**.
- **Sequential Bayesian Causal Filtering:** Variational inference is used to efficiently update the evolving posterior distributions over the entire causal structure.

Expected Advantages:

BTCC is anticipated to deliver reduced catastrophic forgetting (due to stable pathways), stronger forward and backward knowledge transfer (through reusable modules), improved uncertainty calibration, and more robust, dynamic modularity. The long-term vision is to achieve **continual knowledge organization** and cumulative intelligence.